PAX SERVICES GROUP

PAXSEAL™ BUILDING LIFECYCLE PLAN

# Roof Lifecycle Cost Analysis

MARKET PROJECTION
**Typical warehouse**

Culbertson (N. Virginia) market
October 14, 2026

## Planned maintenance is projected to save the owner of a typical Culbertson (N. Virginia) warehouse $869K over the next 20 years.

About 28% less than reactive repair: $539K lower spending plus $330K more roof life at year 20. Range $391K–$1.19M from conservative to upside assumptions; $469K in today's dollars at a 7% discount rate.

| MARKET | PROPERTY | ROOF AREA | SYSTEM | AGE · CONDITION | NEW ROOF TODAY |
|---|---|---|---|---|---|
| Culbertson (N. Virginia) | Warehouse | 100,000 sq ft | TPO | 8 yrs · Good | $900K |

**PAXSEAL NET COST**
**$2.20M**
$2.11M spent + $540K roof used − $450K roof left

**REACTIVE NET COST**
**$3.07M**
$2.65M spent + $540K roof used − $120K roof left

**VALUE PROTECTED**
**$490K**
$32K/yr more NOI at a 6.4% cap rate, if the owner pays for the roof

**ROOF PRICES**
**2× in 15 yrs**
At 4.6%/yr, the long-run trend in commercial roofing costs

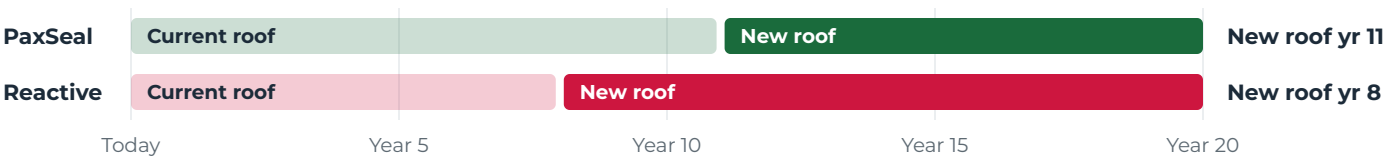
### Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.




### Roof life

When each scenario needs a new roof.



## Where the savings come from

20-year totals

| | PAXSEAL | REACTIVE | SAVINGS |
|---|---|---|---|
| PaxSeal fees | $371K | — | **−$371K** |
| Repairs paid by owner | $219K | $1.29M | **+$1.07M** |
| Expected storm damage | $45K | $68K | **+$23K** |
| Roof replacements | $1.48M | $1.29M | **−$190K** |
| **Cash spending** | **$2.11M** | **$2.65M** | **+$540K** |
| Current roof value used | $540K | $540K | — |
| Roof life left at year 20 | −$450K | −$120K | **+$330K** |
| **Net 20-year cost** | **$2.20M** | **$3.07M** | **+$870K** |

## What this means

1. **A later replacement.** The next new roof moves from year 8 to year 11, leaving 10 years of roof life at year 20 instead of 2.
2. **Smaller repairs.** Emergency repairs run about 3× the planned rate, and the PaxSeal fee covers half of planned repairs. Together that saves $1.07M.
3. **The program pays for itself.** Each $1 of fees avoids about $2.45 of other cash roof costs.

## Local risk profile

Prince William County, VA (51153)

### Thunderstorm wind / derecho, snow and ice

Strong wind and winter weather are both rated Relatively High by FEMA, and tropical storm remnants arrive about once every 12 years. Expected storm damage to this roof over 20 years is **$45K** if maintained and **$68K** with reactive repair only, before insurance. A documented maintenance history also helps when an insurer reviews a claim.

| | |
|---|---|
| Hail | **Relatively Moderate** |
| Strong wind | **Relatively High** |
| Tornado | **Relatively Moderate** |
| Hurricane | **Relatively Moderate** |
| Winter weather | **Relatively High** |
| Ice storm | **Relatively Low** |

FEMA National Risk Index rating

## Key assumptions

The basis for each figure in this analysis.

| ASSUMPTION | VALUE | BASIS |
|---|---|---|
| Roof life | **20 / 15 yrs** | Maintained / reactive. Industry life tables; maintenance adds ~5 yrs (NRC Canada, warranties) |
| Roof cost growth | **4.6%/yr** | BLS Producer Price Index, nonresidential roofing, 2007–2025 |
| Replacement cost | **$9.00/sq ft** | Culbertson (N. Virginia) contractor price ranges |
| Reactive repair cost | **3× planned** | Industry estimates. PaxSeal fee covers half of planned repairs |
| Storm damage | **0.15%/yr** | Of roof cost, before insurance; ×1.5 if reactive. FEMA National Risk Index |
| PaxSeal fee | **$13,800/yr** | Year 1, rising 3%/yr. Estimate; your proposal states the actual fee |
| Cap rate | **6.4%** | Culbertson (N. Virginia) warehouse / industrial market reports |

**Next step:** a roof assessment and a written PaxSeal proposal for this building. Call PAX Services Group at 1-855-729-7510 or email info@paxserv.com.

**Estimates, not a quote.** Projections from Culbertson (N. Virginia) market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20; both roof-value figures are in today's dollars. Storm damage is before insurance. Property value is the yearly gain in net operating income divided by the cap rate, for income property where the owner pays for the roof.